

TS1

$$h[m] = \hat{s}[m] - \hat{s}[m-4] \rightarrow \hat{y}[m] = x[m] - x[m-4]$$

$$(a) x[m] = \cos(\omega_0 m T_s) = \cos(\theta m)$$

$$\hat{y}[m] = \cos(\theta m) - \cos(\theta(m-4)) \rightarrow \hat{y}[m] = A \cos(\omega_0 m T_s + \varphi)$$

Identidades Trigonométricas

$$\cos(A) - \cos(B) = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$A = \theta m$$

$$B = \theta(m-4)$$

$$\frac{A-B}{2} = \frac{\theta m - \theta(m-4)}{2} = 2\theta$$

$$\frac{A+B}{2} = \frac{\theta m + \theta(m-4)}{2} = \theta m - 2\theta$$

$$\hat{y}[m] = -2 \sin(\theta m - 2\theta) \sin(2\theta)$$

$$\hat{y}[m] = 2 \sin(2\theta) [-\sin(\theta m - 2\theta)]$$

$$-\sin(\alpha) = \cos\left(\alpha + \frac{\pi}{2}\right)$$

$$\hat{y}[m] = 2 \sin(2\theta) \cos(\theta m - 2\theta + \frac{\pi}{2})$$

$$\hat{y}[m] = 2 \sin(2\omega_0 T_s) \cos(\omega_0 T_s m - 2\omega_0 T_s + \frac{\pi}{2})$$

$$A = 2 \sin(2\omega_0 T_s)$$

$$\varphi = -2\omega_0 T_s + \frac{\pi}{2}$$

$$b) X[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$u[n] = 1 \quad \text{Si } [n] \geq 0$$

$$u[n] = 0 \quad \text{Si } [n] < 0$$

$$y[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{2}\right)^{n-4} u[n-4]$$

$$n \leq 0 \rightarrow u[n] = 0 \quad u[n-4] = 0$$

$$y[n] = 0$$

$$0 \leq n \leq 3 \rightarrow u[n] = 1 \quad u[n-4] = 0$$

$$y[n] = \left(\frac{1}{2}\right)^n \cdot (1) - 0 = \left(\frac{1}{2}\right)^n$$

$$n \geq 4 \rightarrow u[n] = 1 \quad u[n-4] = 1$$

$$y[n] = \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^{n-4}$$

$$y[n] = \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^{-4}$$

$$y[n] = \left(\frac{1}{2}\right)^n - 16 \left(\frac{1}{2}\right)^n$$

$$y[n] = -15 \left(\frac{1}{2}\right)^n$$

	0	$n < 0$
$y[n]$	$\left(\frac{1}{2}\right)^n$	$0 \leq n \leq 3$
	$-15 \left(\frac{1}{2}\right)^n$	$n \geq 4$

$$y[0] = 1$$

$$y[1] = 0,5$$

$$y[2] = 0,25$$

$$y[3] = 0,125$$

$$y[4] = -0,9375 \rightarrow \text{(Pega un salto a los negativos)}$$

$$y[5] = -0,46875 \rightarrow \text{(Dece exponencialmente hacia '0' por el lado negativo)}$$

$$c) \quad x[m] = u[m+1] - u[m-2]$$

$$y[m] = u[m+1] - u[m-2] - u[m-3] + u[m-6]$$

$$y[-1] = 1$$

$$y[0] = 1$$

$$y[1] = 1$$

$$y[2] = 0$$

$$y[3] = -1$$

$$y[4] = -1$$

$$y[5] = -1$$

$$y[6] = 0$$

$$y[m] = \{1, 1, 1, 0, -1, -1, -1\} \quad m = -1, 0, 1, 2, 3, 4, 5$$

Para otras valores de $m \rightarrow y[m] = 0$

Bonus

DESPLAZAMIENTO TEMPORAL

$$y_B[m] = \{1, 1, 1, 0, -1, -1, -1, 0\} \quad \text{para } m = 0, 1, 2, 3, 4, 5, 6, 7$$

DFT

$$Y_B[k] = \sum_{m=0}^7 y_B[m] \cdot e^{-j\frac{\pi}{4}km}$$

$$Y_B[k] = 1e^{-j0} + 1e^{-j\frac{\pi}{4}k \cdot 1} + 1e^{-j\frac{\pi}{4}k \cdot 2} + 0e^{-j\frac{\pi}{4}k \cdot 3} - 1e^{-j\frac{\pi}{4}k \cdot 4} - 1e^{-j\frac{\pi}{4}k \cdot 5} - 1e^{-j\frac{\pi}{4}k \cdot 6} + 0e^{-j\frac{\pi}{4}k \cdot 7}$$

$$Y_B[k] = 1 + e^{-j\frac{\pi}{4}k} + e^{-j\frac{\pi}{2}k} - e^{-j\pi k} - e^{-j\frac{3\pi}{4}k} - e^{-j\frac{3\pi}{2}k}$$

$$Y_B[k] = (1 - e^{-j\pi k}) + (e^{-j\frac{\pi}{4}k}(1 - e^{-j\pi k})) + (e^{-j\frac{\pi}{2}k}(1 - e^{-j\pi k}))$$

$$Y_B[k] = (1 - e^{-j\pi k})(1 + e^{-j\frac{\pi}{4}k} + e^{-j\frac{\pi}{2}k})$$

$$e^{-j\pi k} = (-1)^k$$

$$1 + e^{-j\frac{\pi}{4}k} + e^{-j\frac{\pi}{2}k} = e^{-j\frac{\pi}{4}k}(e^{j\frac{\pi}{4}k} + 1 + e^{-j\frac{\pi}{4}k}) = e^{-j\frac{\pi}{4}k}(1 + 2\cos(\frac{\pi}{4}k))$$

$$Y_B[k] = (1 - (-1)^k) \cdot e^{-j\frac{\pi}{4}k} (1 + 2\cos(\frac{\pi}{4}k))$$

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import matplotlib.pyplot as plt
import numpy as np

# CONVOLUCIÓN LINEAL (Inciso c)

#  $x[n] = u[n+1] - u[n-2] \rightarrow \{1, 1, 1\}$ 
x = np.array([1, 1, 1])

#  $h[n] = \delta[n] - \delta[n-4] \rightarrow \{1, 0, 0, 0, -1\}$ 
h = np.array([1, 0, 0, 0, -1])

# Convolución lineal  $y[n] = x[n] * h[n]$ 
y = np.convolve(x, h)

# CONFIGURACIÓN DE LA DFT DE 8 PUNTOS (Desplazada)

# Ubicamos la secuencia completa a partir de  $n=0$  y rellenamos con ceros hasta  $N=8$ 
y_desplazada = np.zeros(8)
y_desplazada[: len(y)] = y

# Cálculo de la FFT de 8 puntos
Y = np.fft.fft(y_desplazada)
k_vals = np.arange(8)

# GRAFICACIÓN DE RESULTADOS

fig, axs = plt.subplots(2, 1, figsize=(10, 10))

# Secuencia en el tiempo  $y[n]$ 
axs[0].stem(k_vals, y_desplazada, basefmt=" ")
axs[0].axhline(0, color="black", linewidth=1.2)
axs[0].set_title("Secuencia  $y[n]$  Desplazada a  $n=0$ ")
axs[0].set_xlabel("Muestra ( $n$ )")
axs[0].set_ylabel("Amplitud")
axs[0].set_ylim(-1.5, 1.5)
axs[0].grid(True, linestyle=":", alpha=0.6)

# Magnitud de la DFT
axs[1].stem(k_vals, np.abs(Y), linefmt="C1-", markerfmt="C1o", basefmt=" ")
axs[1].axhline(0, color="black", linewidth=1.2)
axs[1].set_title("Espectro de Magnitud  $|Y[k]|$ ")
axs[1].set_xlabel("Frecuencia Discreta  $k$ ")
axs[1].set_ylabel("Magnitud")
axs[1].set_ylim(0, 5.5)
axs[1].grid(True, linestyle=":", alpha=0.6)

plt.tight_layout()
plt.show()

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